

TrES-3b

R-2687

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_200726 · created 2026-08-31T01:08:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459056.6821 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.056
Transit depth (Rp/Rs)^2	2.18 +/- 1.65 [%]
Orbital Inclination (inc)	80.61 +/- 2.82
Airmass coefficient 1 (a1)	0.053 +/- 0.012
Airmass coefficient 2 (a2)	2.97 +/- 1.75
Scatter in the residuals of the lightcurve fit is	24.02 %
Best Comparison Star	#1 - [428, 478]
Optimal Aperture	1.94
Optimal Annulus	10.96
Transit Duration (day)	0.027 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.056 vs 0.16309 (deviation 0.27 σ)
Duration fit vs published	0.027 d vs 0.0567 d (deviation 1.41 σ)
Mid-time O-C	2.5 min
Depth SNR	2.6
Residual scatter	24.02 %

Light curve

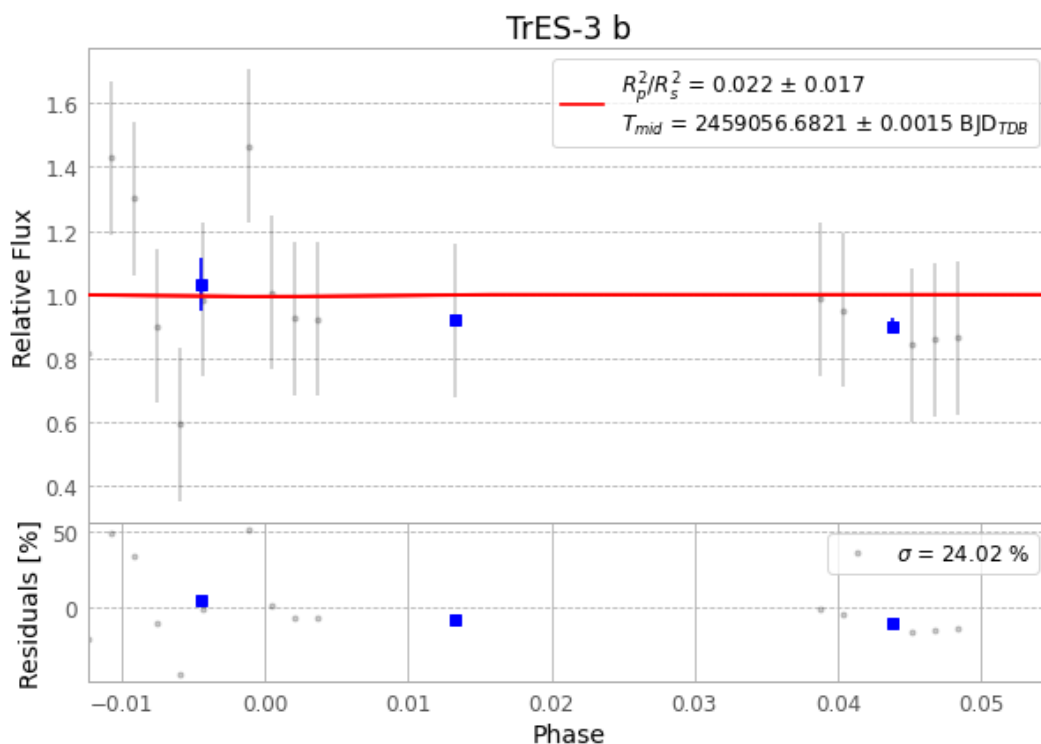


Figure 1 — FinalLightCurve_TrES-3 b_2020-07-25.png (SHA-256 d7ea2f473369e860...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e4fa5c0f4901d984...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

43 FITS frames (28 MB), TRES-3200726035412.FITS → TRES-3200726060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-200726.log (57eb1dab255a...), FinalLightCurve_TrES-3 b_2020-07-25.png (d7ea2f473369...), FinalLightCurve_TrES-3 b_2020-07-25.csv (11c5ab4e483b...)

Compute resources

Wall time 277.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 01:08Z.